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Objectives

- 1 To explain the significance of pricing, both in the economic system as a whole and from a management perspective.
- 2 To explain the context in which pricing decisions are and should be made.
- 3 To relate the concepts and analysis of the previous chapters to more complex and detailed pricing situations.
- 4 To explain the importance of the concept of competitive advantage.

- 5 To explain the concept of value creation and to show its significance in a purchasing model.
- 6 To explain the meaning of market positioning and its strategic implications.
- 7 To discuss market segmentation and targeting strategies.
- 8 To explain the meaning and uses of price discrimination.
- 9 To analyse pricing decisions for firms producing multiple products.
- 10 To analyse pricing decisions for firms producing joint products.
- 11 To explain the concept of transfer pricing and the issues involved.
- 12 To examine the dynamic aspects of pricing, by discussing pricing over the product life-cycle.
- 13 To consider other pricing strategies that firms tend to use in practice.

10.1 Introduction

Pricing is often treated as being the core of managerial economics. There is certainly a fair element of truth in this, since pricing brings together the theories of demand and costs that traditionally represent the main topics within the overall subject area. However, as indicated in various parts of this text, this can lead to an over-narrow view of what managerial economics is about. This chapter will continue to examine pricing in a broader context, but first it is helpful to consider the role of pricing in the economic system.

Price determination is the core of microeconomics; this whole subject area examines how individual components in the economic system interact in markets of various kinds, and how the price system allocates scarce resources in the system. It is a well-established body of theory, with its main elements dating back over a hundred years to the Neoclassical School. Economists may well disagree about how well the market economy works in practice, and this aspect is discussed in the last chapter, but the general framework of analysis regarding the price system is not in serious dispute. Microeconomists tend to focus on pricing at the level of the industry, and on the welfare aspects in terms of resource allocation. As stated in Chapter 2, this perspective on price is what lies behind the assumption that price is the most important variable in the marketing mix. However, although this perspective is important for government policy, it is not the perspective that managers have, or should have, if their objective is profit or shareholder-wealth maximization, or indeed any of the other objectives discussed in Chapter 2.

As far as managers are concerned, price is just one of many decision variables that they have to determine. The majority of managers do not consider it to be the most important of these decision variables, but they do tend to realize the interdependence involved in the decision-making process. Therefore it makes sense at this point to discuss the context of the pricing decision, before focusing on the more detailed analysis of pricing situations. The starting point for this discussion is the concept of competitive advantage.

10.2 Competitive advantage

The concept of competitive advantage was introduced by Porter¹ in 1980, and has been utilized and further developed by many writers on business strategy since then. It provides a very useful means of analysing a firm's success or lack of it in any market. A discussion of competitive advantage is therefore necessary in order to understand the nature of many of the decision variables that are involved in a firm's business strategy, and to put these into a broad perspective. As mentioned in the introduction to the chapter, this involves examining non-price decisions, and explaining the context within which price decisions are or should be made.

10.2.1 Nature of competitive advantage

In order to place the concept in context we initially need to recognize that a firm's profitability depends in general on two factors: market conditions and competitive advantage.

1. *Market conditions.* These relate to external factors, not just for the firm, but also for the industry as a whole. Industries vary considerably in terms of their profitability; thus throughout the 1990s, computer software firms, biotech firms and banks achieved higher than average profitability, while steel firms, coal producers and railways did badly. It is clear that these industry trends can vary from country to country, according to differences in external factors. These external factors are sometimes discussed in terms of another of Porter's concepts, **five-forces** analysis (internal rivalry, entry, substitutes and complements, supplier power and buyer power), or in terms of the **value net** concept of Brandenburger and Nalebuff.² These forces essentially are the ones that were discussed in Chapter 3 as being uncontrollable factors affecting demand, although we should also now be considering the effects on costs. A 1997 study by McGahan and Porter³ concluded that these factors explained about 19 per cent of the variation in profit between firms.

2. *Competitive advantage.* This relates to internal factors, specifically those that determine **a firm's ability to create more value than its competitors**. The study above concluded that these factors explained about 32 per cent of the variation in profit between firms. The concept of **value creation** now needs to be discussed.

10.2.2 Value creation

The value created by a firm can be expressed in the following way:

Value created = perceived benefit to customer – cost of inputs

$$V = B - C \quad (10.1)$$

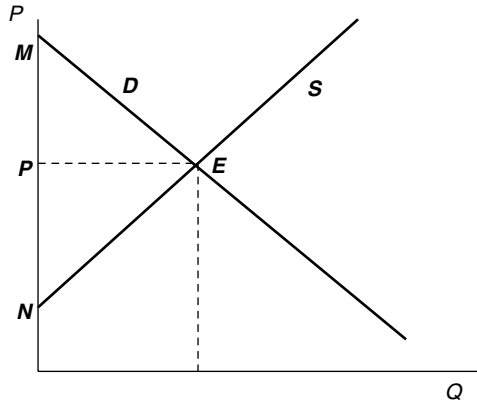


Figure 10.1. Value, consumer surplus and producer surplus.

where V , B and C are expressed in monetary terms per unit. These concepts and their relationships to other important measures can be examined in terms of the graph in Figure 10.1. The demand curve shows the value that consumers place on buying an additional unit of a product, thus indicating perceived benefit in terms of marginal utility, as seen in Chapter 3. The supply curve indicates the marginal cost to firms of producing an additional unit, as seen in Chapter 6. Thus value per unit is measured as the vertical distance between the demand and supply curves, and total value created is measured by the area MEN . The equilibrium price, P , at the intersection of the demand and supply curves can be seen to divide this value into two parts, **consumer surplus** (CS) and **producer surplus** (PS). Consumer surplus is given by $B - P$ in general terms, which, as we have already seen in the last chapter, represents the amount that a consumer is prepared to pay over and above that which they have to pay. Producer surplus is given by $P - C$, which represents supernormal or economic profit. Thus we can write:

$$CS = B - P \quad (10.2)$$

$$PS = P - C \quad (10.3)$$

$$V = CS + PS = (B - P) + (P - C) = B - C \quad (10.4)$$

For the first unit of consumption consumer surplus is given by $M - P$ and producer surplus is given by $P - N$. At the equilibrium point no further value can be created by the firm by producing more output.

In order to make the situation less abstract, consider the following example. A particular consumer may pay £200 for a VCR; they may be prepared to pay £250 for that particular model, even though the marginal cost of production is only £130. In this case the value created by that product item is £120, of which £50 is consumer surplus and £70 is producer surplus.

In order to relate the above analysis to competitive advantage we need to introduce the concept of **consumer surplus parity**. We will assume to start with that consumers have identical tastes, though we will relax this assumption later. When comparing different products and making a purchase decision, rational consumers will try to maximize their total consumer surplus. Firms can be considered as making bids for consumers by offering them more surplus than competitors; a market will be in equilibrium when **each product is offering the same surplus as its rivals**. Thus in the athletic shoe market a particular model of Nike shoe may sell for £10 more than a competitor's product, but if consumers perceive the benefit to be £10 greater, they will obtain the same surplus from consuming each product and therefore be indifferent between the two products. This equilibrium will mean that each product would maintain its market share relative to other products.

If a firm (or product) has a competitive advantage, it is in a position to make more profit than competitors. Firms are able to do this essentially in two different ways: pursuing a **cost advantage** or pursuing a **benefit advantage**. These concepts are examined in more detail shortly but can be briefly summarized here:

- 1 *Cost advantage*. In this case the greater value created ($B - C$) depends on the firm being able to achieve a **lower level of costs than competitors, while maintaining a similar perceived benefit**.
- 2 *Benefit advantage*. In this case the greater value created depends on the firm being able to achieve a **higher perceived benefit than competitors, while maintaining a similar level of costs**.

In order to discuss these issues further we must now consider the assumptions that were made earlier, in Chapter 8 on market structure, regarding the nature of the pricing decision. These included, in particular:

- 1 The firm produces for a single market.
- 2 The firm charges a single price for the product throughout the whole market.
- 3 The firm produces a single product rather than a product line.
- 4 The firm produces a single product rather than a product mix.
- 5 The firm is considered as a single entity, rather than as consisting of different divisions, sometimes in different countries.
- 6 Price is the only instrument in the marketing mix.
- 7 Pricing is considered from a static, single-period viewpoint.

These assumptions are relaxed in the following sections, taking each in turn. The following case study applies the concept of competitive advantage to a very high-profile firm in the telecommunications industry.

Case study 10.1: Mobile phones – Nokia

Emergency Calls⁴

As bad news flows thick and fast, it has become clear that a vicious shake-out is under way in the global telecoms industry, led by the mobile operations on which dozens of companies have staked their futures. Many will want to forget April 24th, a black Tuesday that saw a string of announcements typifying the problems faced by operators and manufacturers alike:

- 1 In Japan, NTT DoCoMo, the wireless arm of NTT, postponed the commercial launch of its third-generation (3G) mobile network from next month until October. It said the system needed further tests before it would be robust enough. The blow is to DoCoMo's prestige rather than its profits, as the firm is still reckoned to be a leader in 3G technology. But the news worried heavily indebted rivals that have looked to DoCoMo to show that 3G can be made to work.
- 2 Germany's Deutsche Telekom announced a first-quarter net loss of €400m (\$369m), and said that it had shifted its focus away from acquiring new mobile customers towards making more money from existing ones. T-Mobile, its wireless arm, made pre-tax profits of €590m, almost 70% more than in the same period last year, and has just won approval to merge with VoiceStream, an American mobile firm. But Telekom remains in trouble. It is shackled with €57 billion of debt and would like to float T-Mobile, but market conditions are too awful.
- 3 Motorola, an American mobile handset and equipment manufacturer, said it was closing its biggest factory in Britain, with the loss of more than 3,000 jobs. It blamed a sudden collapse in demand for mobile telephones and may have to hand back almost £17m (\$25m) in state aid. Other equipment makers are having a torrid time, too. JDS Uniphase, a market leader in fibre optics for high-speed telephony, announced a \$1.3 billion loss for its third quarter and said it would cut its workforce by a fifth.
- 4 Lastly, Ericsson of Sweden, the world's third-biggest maker of mobile handsets, announced a joint mobile-phone venture with Sony. The new business, which will be launched in October, aims to develop new mobile consumer brands and to become a long-term global competitor.

In normal times, Ericsson's deal might have looked like the only piece of good news on an otherwise miserable day. But it came almost immediately after the company had announced a big retrenchment and 12,000 job losses. In the first quarter of this year, it made a SKr4.9 billion (\$502m) pre-tax loss, and it admitted that its mobile handset business is looking shaky. Its shares are worth around 70% less than they were a year ago, and have fallen by more than one-third since the beginning of March.

So it is hardly surprising that Kurt Hellstrom, Ericsson's chief executive, looked subdued at the press conference to announce the tie-up with Sony. He is paying a heavy price for having missed the strategic shift as mobile phones became trendy consumer goods rather than purely functional items. Ericsson underinvested in design and has been eclipsed by niftier rivals, notably Nokia of Finland.

Thus, the deal with Sony is better seen as a measure of how quickly Ericsson has fallen from grace. It has conceded 50% of the venture to the Japanese, even though it is far bigger, selling 43m phones last year against Sony's 7.5m. In an industry that moves quickly, the new venture will produce its first products only some time next year. Mr Hellstrom might no longer be the boss by then.

All over the industry there are clear signs that the telecoms boom has ended. Dramatic retrenchment by top-notch competitors may look panicky, but the truth is that most have little alternative. As the mobile sector has matured, its growth has inevitably begun to slow, leading investors to question the prospects of traditional operators that have placed huge bets on mobile technologies. Not only have capital markets become choosier about telecoms-related financings, but weak equity markets have made it almost impossible to float off mobile ventures and pay back debt. The knock-on effect for mobile-handset manufacturers has become all too evident – not least because they have had to provide billions of dollars in 'vendor financing' so that mobile-network operators can continue to buy their products.

A rare bright spot has been Nokia, the world's biggest mobile maker. Total sales in 2000 increased by 54% over the previous year, to around €30 billion. Net profit also increased by 53%. Even in the turbulent first quarter of 2001 total sales increased 22%, while operating profit rose 8%. These

first-quarter results were better than expected and its share of the global handset market is edging up towards 40% as Ericsson's falls. So are its shares doing well? Relatively, yes: at the end of April they were a mere 40% below their level of a year before.

Nokia succumbs⁵

Then on June 12th there was proof, if proof were still needed, that no technology company is bulletproof. Nokia, the world's leading manufacturer of mobile telephones, gave warning that its second-quarter profits would be lower than expected, and that annual sales growth, previously forecast at 20%, would in fact be less than 10%. Nokia's shares fell by 23%, though they later recovered slightly. Other telecoms firms' share prices suffered too, with the exception of BT, whose shares rose after news of a 3G network-sharing deal with Deutsche Telekom.

Nokia's announcement was portrayed by Jorma Ollila, the firm's boss, as an indication that the slowdown in the American economy is having knock-on effects in Europe. But this explanation is a red herring, says Mark Davies Jones, an analyst at Schroder Salomon Smith Barney. Although Nokia has been slightly affected by falling consumer demand in America, the real cause of its problems is that the market for handsets, which account for nearly three-quarters of its sales, is saturated.

The problem is that people are neither buying new phones, nor upgrading their old ones, as often as they used to. In part this is because network operators, most of which are struggling with huge debts, feel less inclined to subsidise handsets. But it is also because there is no compelling reason to upgrade your phone once it is small and sexy enough. The days of double-digit growth in handset sales are over: the number of handsets sold worldwide – 400m last year, of which 32% were Nokia's – is not expected to rise this year. Any sales growth at Nokia this year will come from increasing its market share. In the UK

there are now over 40 million mobile phone subscribers, and the market penetration rate is around 70%, one of the highest in the world.

What now? The industry has a plan, which is to introduce new mobile-data services. Operators will benefit by being able to charge for these services (since revenues from voice traffic have stopped growing) and handset manufacturers will be able to sell everybody new phones. The problem is that the first incarnation of mobile data, Wireless Application Protocol (WAP) services, was a flop. So the industry's hopes are now pinned on a new technology, General Packet Radio System (GPRS), which is faster than WAP and offers 'always on' connections.

Nokia has extensive 3G contracts in the UK, including three-year agreements with Orange to deliver the radio-access network from their 3G network, and a three-year agreement with Hutchison to deliver a complete range of 3G mobile network infrastructure worth around €500 million.

On June 13th, the GSM Association, an industry standards-setting body, announced a scheme called the 'M-Services Initiative', which defines a standard way to offer graphics and other multimedia content on GPRS phones. The idea is that these new features will encourage users to upgrade their handsets, and thus plug the gap before the arrival of 3G phones in a couple of years' time. The big operators and manufacturers, including Nokia, are backing the scheme, and the first handsets sporting graphics should be in the shops by Christmas. One way or another, this week could prove to be a turning point for the industry.

Questions

- 1 How would you describe Nokia's competitive advantage?
- 2 Explain the implications of the saturation of the handset market for Nokia's competitive advantage, making strategy recommendations.

10.3 Market positioning, segmentation and targeting

Positioning in the market is the most fundamental aspect of a firm's marketing strategy; it precedes any consideration of the marketing mix discussed in Chapter 3. A firm must begin by examining its resources and capabilities relative to the business that it is in, or any business that it could be in. These

resources and capabilities include things like management know-how, technical expertise, reputation and image, patents and trademarks, organizational culture, ownership of specialist resources, relationships with customers and suppliers and location. It should be noted that these factors tend to be ingrained within the organization, meaning that they are not dependent on the presence of certain individuals. Another important characteristic is that they are not easily duplicated. If there is a close match between these firm-specific factors and the key success factors in the industry then the firm may be able to obtain a competitive advantage in either cost or benefit terms.

10.3.1 *Cost advantage*

In basic terms this means achieving a lower level of C while maintaining a level of B that is comparable to competitors. Examples of firms that have pursued this strategy are Woolworth, Wal-Mart, Asda (now owned by Wal-Mart) and McDonald's. In some cases the firm claims to have the same quality as competitors at a lower cost, while in other cases the firm's perceived quality may be lower, but the cost and price substantially lower. In this situation we no longer need to assume that consumers have identical tastes; the firm can provide at least as much consumer surplus to some customers as competitors, while maintaining at least the same profit. The determination of how the value created should be divided into consumer surplus and producer surplus involves pricing strategy, and this is considered in subsection 10.3.3.

Cost advantage can be achieved in a number of ways: economies of scale and scope, the learning curve, production efficiency, control over inputs, transaction efficiencies and favourable government policies may all play a part. We will take the car industry as an overall example. In the case study on Nissan in Chapter 6, it was seen that transactional efficiencies in terms of dealing with suppliers formed an important component of a cost reduction strategy, as did increasing capacity utilization. When BMW took over Rover in the UK they were hoping, among other things, to obtain favourable government treatment in terms of grants and subsidies. Other car firms are concentrating on producing a smaller number of basic platforms for their different models in order to gain more economies of scale. Korean car manufacturers, like Hyundai and Daewoo, have concentrated on producing no-frills cars of lower perceived quality, but at a much lower price than other cars. The British sports car manufacturer TVR produces high-performance cars at a lower price than cars of similar performance levels produced by other manufacturers; this is achieved by simplifying the production process, using components from other cars, omitting some high-tech gadgetry and cutting back on marketing overheads.

10.3.2 *Benefit advantage*

A strategy aimed at achieving a benefit advantage involves offering a product with a higher level of B than competitors while maintaining a similar level of C .

Examples of firms in the car industry that have pursued this type of strategy are Honda, BMW (particularly in the 3 series) and Toyota with its Lexus model, although with this Toyota originally pursued both ends of the spectrum by producing a car at a lower price than the luxurious Mercedes and BMW saloons, but which used and provided more in the way of new technology. Other companies aiming at a benefit advantage may price their products somewhat higher than competitors but boast a significantly greater quality. Porsche is an example. Some companies, like Aston Martin and Ferrari, take this to the extreme; they charge very high prices, but promise the very best in quality.

Just as with cost advantage, benefit advantage can be achieved in a number of ways, even in a given industry. Reputation or marque counts for a large amount in the luxury car market. The Japanese manufacturers in general have tried to make up for a lack of this characteristic by using and providing more advanced technology, such as ABS, traction control, variable valve timing, four-wheel steering, climate control, satellite navigation and other gadgets. Aston Martin emphasizes build quality and customer service. BMW stresses 'German' engineering and reliability. It should be emphasized however that firms pursuing a benefit advantage do not have to be producing a luxury product; the quality of the product does not have to be high in general, it simply has to be perceived as being higher than competitors in that particular market segment. For example, the Japanese manufacturers mentioned earlier have mostly been producing cars in the medium or low price range. Again, as with cost advantage, the firm must determine how the value created should be divided into consumer and producer surplus. This means determining a pricing strategy. The general considerations regarding this are discussed in the next subsection.

10.3.3 *Competitive advantage, price elasticity and pricing strategy*

We now need to apply the concept of price elasticity of demand (PED), examined in Chapter 3, to the concept of competitive advantage in order to see how firms should generally price their products. The more detailed and quantitative aspects of pricing strategy will be considered in later sections of the chapter.

a. Cost advantage

There are two possible strategies for firms with a cost advantage:

- 1 The greater value that it creates can be given over to customers in the form of consumer surplus by charging a lower price, with the firm maintaining the same profit margin as competitors; in this case the firm should increase its market share and thus make more total profit.
- 2 The greater value created can be translated into producer surplus, meaning a greater profit margin, by charging the same price as competitors and providing the same perceived quality.

Table 10.1. Price elasticity and competitive advantage

	High PED	Low PED
Cost advantage	Cut price Gain market share	Maintain price Gain profit margin
Benefit advantage	Maintain price Gain market share	Increase price Gain profit margin

Choice of the appropriate strategy depends on the PED of the product. If PED is high, a lower price will allow the firm to capture significant market share from competitors. On the other hand, if PED is low, large price cuts will not increase market share much; in this case it is better to keep price at the same level as competitors and increase profit margin; see Table 10.1.

b. Benefit advantage

Again there are two possible strategies:

- 1 The greater value created can be given over to customers in the form of a larger consumer surplus by charging the same price as competitors and translating the greater consumer surplus offered into greater market share.
- 2 The greater value created can be translated into producer surplus by charging a higher price than competitors, while still maintaining the same consumer surplus. This will increase profit margin.

In this situation of benefit advantage, if PED is high it is better to maintain price at the same level as competitors and gain market share. If PED is low then it is better to increase price, since this will not affect market share much, and concentrate on increasing profit margin; again see Table 10.1.

10.3.4 Segmentation and targeting

These two aspects of strategy are closely related to competitive advantage and positioning. A **market segment** refers to **a group of consumers in a wider market who are homogeneous in some respect related to their buying behaviour**. Markets can be segmented in many ways according to different characteristics: demographic factors (income, age, ethnicity and sex, for example), location, frequency of purchase or use, lifestyle and newspaper readership are all common means of segmentation.

Why is it useful for firms to segment markets? Different segments have different tastes and therefore respond differently to the various marketing mix variables. A high-income segment, for example, may have a lower PED for a product than a lower income segment, meaning that they may be less price-sensitive; on the other hand, they may be more quality-sensitive. A firm may therefore want to select target markets according to its competitive advantage.

However, strategy selection does not always occur in this order. For example, a firm may observe that a certain market segment is not currently being adequately catered for in the market; it may therefore pursue a competitive advantage in producing some product or range of products that appeal to that segment. In the first case we could say that the firm is **supply-driven**, while in the second case the firm is **demand-driven**.

Targeting strategies are often classified into two main categories: **broad coverage** strategies and **focus** strategies.

a. Broad coverage strategies

These strategies aim to **provide for the needs of all or most of the segments in the market**. This can be achieved either by providing a general product line of a 'one-size-fits-all' type, or by providing a product line consisting of a variety of related products that are more-or-less customized for each segment. An example of the first approach is the strategy used by most commercial banks for attracting bank depositors, at least until recently. Little distinction was made according to age, income, occupation or frequency of cheque-writing. The main attraction of this approach is that it allows more standardization, and therefore leads to greater economies of scale.

However, most firms try to appeal to different segments by producing different products in a product line. For example, Nike produces athletic shoes in a huge variety, according to type of usage, frequency of usage, weight and build of user, sex of user, aesthetic preference of user and so on; prices therefore vary considerably, from about £20 a pair to over £100. Mercedes has recently reinvented itself in this direction; having originally specialized in producing luxury saloons, it is now selling sports cars, the economy 'A' class, and SUVs, with a price range from £15,000 to over £100,000. With this type of strategy economies of scale are less important, while economies of scope may be more important. This is not to say that cost advantage is always relevant here; benefit advantage may be more important, at least in certain market segments.

b. Focus strategies

These involve a firm either **offering a single product, or serving a single market segment, or both**. The first strategy may apply in markets where there is a lot of product homogeneity, like steel production, or it may apply to small specialist firms, particularly in industrial markets. More commonly a firm will specialize in providing for a single market segment; TVR specializes in producing high-performance sports cars, as do Lotus and Marcos. Bose specializes in producing high-quality speaker systems. Tag Heuer specializes in producing high-quality watches, particularly for sporting people. However, the focus does not have to be on high quality, associated with benefit advantage. Firms like Asda, K-mart, McDonald's and Woolworth have specialized in providing goods of 'adequate' quality at a low price. In this case, economies of scale and other cost reduction factors are of vital importance in yielding a cost advantage.

10.3.5 Role of pricing in managerial decision-making

We can now see that pricing is best considered as one of many interdependent decisions that firms have to make, and often is not the most fundamental. Firms need to start from their objectives in general terms, and then consider their competitive advantage in terms of their resources and capabilities, in conjunction with the external conditions of the market in terms of customers and competitors. Normally (with exceptions examined later in this chapter) this means starting with a positioning decision, leading to a product decision, and then considering the other elements of the marketing mix. Pricing is considered in this context, and affects the buying decisions of consumers through determining the amount of consumer surplus created.

Firms can pursue either a cost advantage or a benefit advantage. In either case the firm has to create more value ($B - C$) than competitors. A firm then must determine a pricing policy to divide this value between consumer surplus and producer surplus. If a firm wants to increase market share, it has to create more consumer surplus ($B - P$) than its competitors. An alternative strategy is for the firm to increase profit margin by creating more producer surplus ($P - C$).

It can be seen, therefore, that the main priority of management is to identify and determine products where value creation is possible. Only then is the division of such value into consumer and producer surplus considered. Thus pricing is usually seen as being a secondary consideration.

Some of the more complex methods of increasing consumer surplus through pricing strategy are considered at the end of the chapter in section 10.7. First of all, the main techniques of pricing must be considered, based on the methodology in Chapter 8, but relaxing the other assumptions listed earlier.

However, before moving on to techniques of pricing, the concepts of positioning, segmentation and targeting are examined in the context of the handheld computer industry.

Case study 10.2: Handheld Computers – Palm

One Palm flapping⁶

Carl Yankowski is having a hard time turning Palm's promise into profits

By rights, it should not have been this hard. Carl Yankowski, chief executive of Palm, the leading maker of handheld computers, is a former president of Sony's American operations. He is a big, dressed-in-black 'gadget guy' with just the right combination of consumer-marketing experience and technology savvy to straddle the gap between computing and consumer electronics. From its start in 1996, Palm single-handedly generated the personal digital assistant (PDA) craze, succeeding where many had

failed before, and it is now leading a drive into the new wireless world. Computing meets telecoms meets consumer electronics – the palmtop becomes the new desktop. And here, or so it might seem, in the right place at the right time, is Mr Yankowski.

Yet he is struggling. For all Palm's success in defining and leading a booming new industry – its operating system runs more than three-quarters of the world's PDAs, giving it an almost Microsoft-like monopoly – the company itself is in a very un-Microsoft-like financial mess. Recently, it issued a warning that its losses in the current quarter would be twice what had been expected – as much as \$190m. Its sinking fortunes also scuttled a planned merger

with Extended Systems, a deal that was meant to take it to new heights in the corporate market. Palm's shares have fallen by 90% over the past six months.

The company's difficulties do not all lie at the feet of Mr Yankowski – he took over only a year and a half ago, as Palm was freeing itself from 3Com, its former corporate parent and the source of its risk-averse corporate culture. But, like it or not, they are now his to solve.

The biggest problem is that Palm is having a hard time finding the right strategy. At the moment, it follows a combination of Microsoft's with Apple's that ends up weaker than either. Like Apple, it makes its own hardware and software: a line of PDAs and the famed Palm operating system (OS) that is the secret of its success. Like Microsoft, it also licenses its OS to other companies, ranging from Handspring, which was started by Palm's original founders, to Sony and several mobile-phone makers.

The downside to this is that Palm's licensees have proved all too good at making hardware. Today, a mere two years after it released its first PDA, Handspring is beating Palm in sales. And both Sony and Handspring have pushed their hardware further than Palm, introducing innovations such as expansion slots and add-on devices from phones to music players. The result is that the PDA business is quickly taking on the savage character of the PC industry, with commodity products, falling margins and cut-throat competition.

Mr Yankowski inherited most of this, and it had too much momentum for him to change it quickly or easily. If Palm stops licensing its operating system, it risks losing out to OS competitors such as Microsoft and Psion. If it stops making hardware entirely, it would take the best-known brand of PDA out of circulation.

The enthusiasm over Mr Yankowski's arrival in late 1999 was based largely on his background, which suggested that there might be a third way for the company. Aside from his Sony experience, which placed him at the heart of the best consumer-electronics firm just as it was embracing digital technology, his career has been a virtual tour of great marketing firms: Reebok, General Electric, Pepsi, Memorex and Procter & Gamble. Starting it all was an engineering degree from MIT.

The hope was that Mr Yankowski could combine Sony's design and marketing skills with Palm's technology. His type of consumer-marketing

experience can make the difference between a niche gadget and a Walkman-like hit. Which is why it is so puzzling that Palm has changed so little since his arrival.

CORPORATE PRIORITY

Many expected him to push the company faster into the consumer market, with brightly-coloured PDAs and extra consumer features such as MP3 and video. Instead, he has made his main priority the staid corporate market. At present, most Palms make it into the office thanks to somebody's personal expense account. Mr Yankowski's aim is to encourage IT managers to purchase them directly for employees, much as they buy PCs.

This is not a bad strategy – the corporate PDA market is about the same size as the consumer market, and both have lots of potential – but it may be a waste of Mr Yankowski's special talents. While he tries to sell his firm's strait-laced productivity tools, in black and grey, to corporate purchasing managers, his old company, Sony, is generating enviable buzz with a cool purple PDA that plays videos and has a headphone jack. Worse, the corporate market is the one in which Palm faces its toughest competition, in the form of Research in Motion's Blackberry interactive pagers, which have generated Sony-like excitement among the suits.

Palm's recent results have at last provoked Mr Yankowski into thinking more broadly. He is now in management retreats with his staff and is expected to announce a new strategy soon. But his options get more limited by the day. Palm's finances are too rocky to get into a consumer-marketing race with Sony. Nor does it have the products to justify that. The first Palms designed on Mr Yankowski's watch are now out. They do little more than add an expansion slot like the one Handspring has had for two years.

Meanwhile, the collapse of the planned merger with Extended Systems, which has had success selling to IT managers, limits the push into the corporate sector. And abandoning hardware entirely would reduce Palm to a software and services firm – hardly the place for a consumer-marketing guru. Mr Yankowski does not have much more time to find the right answer.

Questions

- 1 Describe Palm's positioning in the market, in terms of resources, capabilities, cost and benefit advantage.
- 2 What criteria can Palm use to segment its market?
- 3 Evaluate Palm's targeting strategy.

10.4 Price discrimination

10.4.1 Definition and conditions

a. Definition

Price discrimination has been defined in a number of different ways. The simplest definition relates to the situation where a firm sells the same product at different prices to different customers. However, the most useful definition involves a firm **selling the same or similar products at different prices in different markets, where such price differentials are not based on differences in marginal cost**. This latter definition covers a broader range of situations and leads to a greater understanding regarding the nature of price discrimination. Using this definition we can consider the following common practices as price discrimination:

- Airlines that charge different fares for the same journey at different seasons of the year.
- Universities that charge higher fees to overseas students than to home students.
- Restaurants that offer 'early bird' dinners.
- Professional people, like doctors, accountants and consultants, who charge richer clients higher fees than poorer clients for the same service.
- Exporters who charge lower prices abroad than they charge domestically.
- Prestige outlets that charge higher prices than high street or discount stores for the same products.
- Health clubs that sell off-peak memberships.
- Supermarkets that offer points or reward schemes to regular shoppers.
- Happy hour.

We can now consider the conditions that are necessary for price discrimination to be possible.

b. Conditions

There are two fundamental conditions:

- 1 There must be different market segments with different demand elasticities.
- 2 The market segments must be separated so that there is no possibility of resale from one segment to another.

In order to illustrate the importance and application of these conditions let us consider the example of a butcher. Instead of advertising his meat prices he may wait until the customer comes into the shop and charge prices according to an assessment of the individual customer's ability and willingness to pay. Such a strategy may be successful in the case of customers who are ignorant of the prices being charged to other customers; however, once they become aware of this situation, and assuming that other butchers are following the

same practice, high-paying consumers will find ways to avoid paying higher prices by buying their meat from other customers, in other words **arbitrage** will occur. Arbitrage refers to the practice of **buying at a lower price and selling at a higher price to make a riskless profit**. Thus in this instance a butcher cannot successfully practise price discrimination. On the other hand, a dentist is in a position to do so; patients cannot ask other patients to have their teeth capped for them. Thus price discrimination is generally easier in the markets for services, especially personal and professional services.

At this point one might ask why a firm would want to practise price discrimination. Essentially it is a method of reducing the amount of consumer surplus and transferring it to the producer in terms of additional profit. This aspect is most easily seen by considering different types of price discrimination.

10.4.2 Types of price discrimination

There are different ways of classifying these types of discrimination. We will discuss the degree of price discrimination first, since it leads to a better understanding of the resulting impact on profit; then bases for price discrimination will be considered, in terms of different ways of segmenting markets.

a. Degree of price discrimination

Economists tend to classify price discrimination into three main categories, according to the extent to which consumer surplus is transferred to the producer.

1. First-degree price discrimination. This is sometimes referred to as perfect price discrimination, since all the consumer surplus is transferred to the producer. For this to be possible the producer must have complete knowledge of its demand curve, in terms of knowing the highest price that each customer is prepared to pay for each unit, and be able to sell the product accordingly. This situation is extremely rare, but an example is an auction market, like the Treasury Bill market. Figure 10.2 illustrates the situation. It is assumed in this case that marginal costs are constant, for simplicity. The whole consumer surplus, GCP_3 , becomes revenue and profit for the producer.

2. Second-degree price discrimination. In this situation a firm may charge different prices for different levels of consumption. The first Q_1 units may be sold at the high price of P_1 ; the next block of units, $Q_2 - Q_1$, may be sold at the price P_2 , and the last block of units, $Q_3 - Q_2$, may be sold at the price of P_3 . This situation is also shown in Figure 10.2 and the size of the consumer surplus is given by the three triangles, GAP_1 , ABE and BCF . This type of strategy may be used in the selling of a product like electricity.

In other situations a reverse type of strategy is sometimes used, where early buyers pay a lower price and later buyers have to pay more. Examples are the

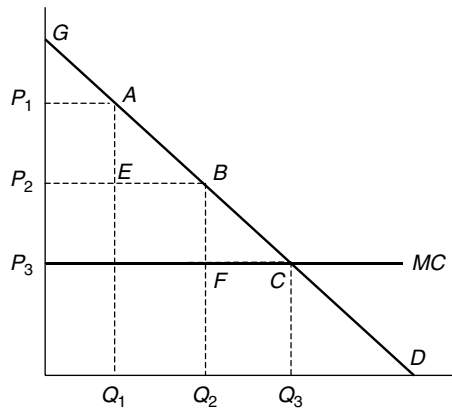


Figure 10.2. Degrees of price discrimination.

selling of tickets to certain events, like concerts, and the selling of certain types of club membership. Usually buying early involves more inconvenience or inflexibility for the buyer, thus segmenting the market.

Second-degree price discrimination often features a two-part tariff or pricing system. There is some lump sum charge and then a user charge. Examples are cellular phone and cable television providers, car rentals, and various types of club membership. In the first two cases the lump sum is minimal and the user fees relatively large; with cable for example, the boxes are heavily subsidized by the provider and are often supplied free of charge, but if a second box is required, the buyer has to pay around £300. This also explains the apparent anomaly that replacement blades for many razors cost more to buy than whole disposable razors. On the other hand, with many club memberships the lump sum charge, or entry fee, is relatively large, while user fees are nominal.

3. Third-degree price discrimination. This is, in practice, the most common form, where a firm divides its market into segments and charges different prices accordingly. Markets can be segmented in various ways for this purpose, and this aspect is now discussed.

b. Segmentation for price discrimination

Firms can segment markets on different bases; the following are the most common.

1. Time. Many products that have seasonal demand have different prices at different time periods; airlines, hotels, restaurants, cinemas, power providers and many other industries have this feature. Demand is less elastic at peak season, so prices are higher then.

2. Location. In some cases the price of a product may be higher in one location than another because of a higher transportation or other cost. This

is not price discrimination. However, when the price in one location is higher than is justified by a difference in marginal cost, because of demand being less elastic, then this does involve discrimination. It should also be mentioned that sometimes a kind of reverse price discrimination occurs. Instead of charging different prices when marginal cost is the same, some firms will charge the same price even when marginal costs are different. This occurs, for example, when there are differences in transportation costs between different markets, but a universal price is charged. Again this could be justified by differences in demand elasticities between different markets, if demand is more elastic in those areas with higher transportation costs.

3. Product form. Many firms offer a product line where the top-of-the-line model is considerably more expensive than the other products in the line. This may be true even when there is little difference in the cost of production of the different products. In this case there may be some consumers who are determined to have the best (stereo, television, car, mountain bike, tennis racquet), and therefore, again, their demand is less elastic in the normal price range.

4. Customer. This is the most controversial type of price discrimination because of concerns regarding fairness. Recalling the discussion in Chapter 1, fairness is a normative issue and is not relevant to the present analysis. Unfortunately the term 'discrimination' is an emotive word and tends to be associated with normative issues, but it is not our purpose to discuss these here. It is simply necessary to point out that firms can increase their profit by charging higher prices to customers whose demand is less elastic, provided that there is no possibility of resale between market segments, as explained earlier. This explains the discrimination carried out by many providers of professional services, including universities. It also explains the price discrimination practised by many exporters, charging higher prices in the domestic market in spite of lower costs; it is worth noting that in this case there are limited opportunities for arbitrage, which is why many UK residents are now buying expensive consumer durables like cars in mainland Europe and shipping them back to the UK. This aspect is discussed in more general terms in the next subsection.

10.4.3 Price discrimination in the European Union

The recent introduction of the single currency in the euro zone has had very important implications for the practice of price discrimination in the EU. This has always existed on the basis of both customer and location, but the practice has been obscured in the past by the fact that prices were quoted in different currencies. Now that the euro is in circulation such discrimination is

much more transparent, as prices in different countries are directly comparable. *L'Expansion*, a French magazine, conducted a survey of the euro zone in early 2001 and found that huge discrepancies existed for many products. For example, a kilo of beef cost €9.90 in Madrid, €15 in Paris, and €21 in Amsterdam; a 5-kilo pack of detergent cost €9.80 in Brussels but €24.30 in Helsinki; a packet of proprietary aspirin cost €3.70 in Athens but €12.90 in Rome and Berlin.

The introduction of the euro has therefore caused many multinational firms to re-evaluate their pricing strategies. Some firms, like *The Economist* magazine, have taken to charging a universal single price in all countries, €4.35, which is approximately the average of its previous prices. It has adopted this strategy on the basis that 'it was better to send a consistent price signal to customers than to price by market'.⁷ While this may be true, it could also be claimed that, in a situation of price transparency, the possibility of resale from one market segment to another meant that the second condition for price discrimination to operate successfully was not applicable, and therefore *The Economist* really had no choice. **Parallel importing**, the practice of **buying more cheaply in one country and selling at a higher price in another country**, is a form of arbitrage and has become very profitable within the European Union. It is particularly common in the pharmaceutical industry, where price differentials are very large; it has been estimated that parallel imports of drugs are worth £700 million annually in the UK alone.

Other firms are not necessarily in the same situation as *The Economist*, since they often sell differentiated products in different markets. For example Mandarin Duck is an Italian company that makes fashionable handbags and accessories. In the past it has practised price discrimination based on different standard mark-ups for fashion accessories in different countries; in Italy retailers operate on mark-ups of about 2.15 times the wholesale price, while in France the multiple is 2.29, and in Germany the multiple is 2.5. Although its prices in different countries have converged with the introduction of the euro, the firm now concentrates on tailoring its products to suit particular markets, selling more expensive items in the wealthier markets and cheaper ranges in the less wealthy areas.

However, even with the increased transparency brought about by the single currency, it will still be difficult to compare the true prices charged by manufacturers and retailers. This is because of the varied systems of incentives that are often offered by suppliers, involving rebates based on annual sales volumes and subsidized credit terms. A. T. Kearney, a management consultancy firm, recently found 216 different pricing structures in contractual terms between buyers and suppliers in Europe's main consumer-goods markets. Given this situation it is likely that a considerable amount of price discrimination will persist even after the introduction of the single currency.

Having discussed the different degrees of price discrimination, bases for segmenting markets, and some practical examples, we can now move on to a more detailed quantitative analysis of the situation.

10.4.4 Analysis

Consider the following situation:

SP10.1 Price discrimination

Valair is an airline flying a particular route that has seasonal demand. The firm's total demand is given by:

$$Q = 600 - 4P \quad (10.5)$$

where Q is the number of passengers per year, in thousands, and P is the fare (in £). In the peak season the demand is given by:

$$Q_H = 320 - 1.5P_H \quad (10.6)$$

and in the off-season the demand is given by:

$$Q_L = 280 - 2.5P_L \quad (10.7)$$

assume that fixed costs are £6 million per year and that marginal costs are constant at £60 per passenger. Thus the cost function is given by:

$$C = 6000 + 60Q \quad (10.8)$$

where C is total costs (in £'000).

- Calculate the profit-maximizing price and output without price discrimination, and the size of the profit.
- Calculate the profit-maximizing price and output with price discrimination, and the size of the profit.
- Calculate the demand elasticities of the two segments at their profit-maximizing prices.

Solution

- Without price discrimination*

Reviewing the procedure described in Chapter 8:

$$\begin{aligned} 1 \quad P &= (600 - Q)/4 \\ R &= (600Q - Q^2)/4 \end{aligned}$$

$$2 \quad MR = (600 - 2Q)/4$$

$$\begin{aligned} 3 \quad MR &= MC \\ (600 - 2Q)/4 &= 60 \\ 600 - 2Q &= 240; 2Q = 360 \end{aligned}$$

$Q = 180$, or **180,000 passengers per year**

$$4 \quad P = (600 - 180)/4 = \textbf{£105}$$

The profit is given by $R - C$:

$$\begin{aligned} &= 105(180,000) - [6,000 + 60(180)]1,000 \\ &= 18,900,000 - 16,800,000 \\ &= \textbf{£2,100,000} \end{aligned}$$

b. *With price discrimination*

We now examine each segment in turn:

Peak segment (H)

$$1 \quad P = (320 - Q)/1.5$$

$$R = (320Q - Q^2)/1.5$$

$$2 \quad MR = (320 - 2Q)/1.5$$

$$3 \quad MR = MC$$

$$(320 - 2Q)/1.5 = 60$$

$$320 - 2Q = 90$$

$$2Q = 230$$

$$Q = 115, \text{ or } \textbf{115,000 passengers per year}$$

$$4 \quad P = (320 - 115)/1.5 = \textbf{£136.67}$$

Off-peak segment (L)

$$1 \quad P = (280 - Q)/2.5$$

$$R = (280Q - Q^2)/2.5$$

$$2 \quad MR = (280 - 2Q)/2.5$$

$$3 \quad MR = MC$$

$$(280 - 2Q)/2.5 = 60$$

$$280 - 2Q = 150$$

$$Q = 65, \text{ or } \textbf{65,000 passengers per year}$$

$$4 \quad P = (280 - 65)/2.5 = \textbf{£86}$$

In order to obtain the size of the profit it is necessary to calculate total revenue and subtract total costs. Note that it is incorrect to compute the profits in each segment separately and then add them together. This would double-count the fixed costs.

Total revenue = $R_H + R_L = 136.67(115,000) + 86(65,000) = £21,307,050$

Total costs = $[6,000 + 60(115 + 60)]1,000 = £16,800,000$

Profit = **£4,507,050**

- c. The demand elasticities in each segment can also be obtained using the point elasticity formula:

$$\text{Peak: PED} = -1.5(136.67/115) = -\mathbf{1.783}$$

$$\text{Off-peak: PED} = -2.5(86/65) = -\mathbf{3.308}$$

At this point a number of general conclusions can be drawn from comparing the situation with price discrimination and the situation without price discrimination:

- 1 Total output is the same in both situations (note that this is not true if the cost function is non-linear).
- 2 The prices with discrimination 'straddle' the price without discrimination, meaning that one is higher and the other is lower. If there are more than two market segments one or more prices will always be higher and one or more prices will always be lower.
- 3 The segment with the higher price will have less elastic demand and vice versa.
- 4 Profit is always higher under price discrimination. This is because some of the consumer surplus is transferred to producer surplus, as seen earlier.

The following case study examines price discrimination in the airline industry, concentrating on bases for discrimination and the importance of differences in price elasticity of demand.

Case study 10.3: Airlines

Britain takes to the air⁸

Low-cost carriers are transforming not just the travel business in Britain, but also the way people live

The air of gloom surrounding much of European business made Ryanair's results, announced on June 25th, particularly impressive. The low-cost airline reported a 37% year-on-year increase in pre-tax profits, and its chief executive, Michael O'Leary, said he expects business to grow by 25% over the next year.

Thanks to Ryanair and its sort, Britons are beginning to hop on and off planes the way Americans do.

Air travel in and around Britain has grown by nearly 40% in the past five years, but the really spectacular growth has come from the low-fare airlines, which have carried around 20m passengers in the past year. By spotting and satisfying the untapped demand for travel from and between the regions, they have fuelled the growth of Britain's smaller airports and undermined Heathrow's dominance.

EasyJet, the first of the low-cost carriers, was set up in 1995 at Luton. Eastwards around the M25 at Stansted are Ryanair, Go, the low-cost offshoot of British Airways (BA) sold to a management buy-out earlier this year, and Buzz, the British arm of KLM, which uses the airline partly to feed its international hub at Amsterdam. While Heathrow has seen the number of passengers rise by about 19% over that period, traffic at Luton and Stansted has more than trebled. Traffic at Liverpool's airport nearly quadrupled.

Demand for air travel is highly elastic. Bring down the price and sales rise sharply. The low-fare carriers are often cheaper not just than the mainstream operators but also than the railways. While low-fare airlines keep their fixed costs to a minimum, the railways are burdened by the need to maintain and improve their crumbling network. Last year was a disaster for them. A crash blamed on a cracked rail led to mass disruption as managers tried to locate and mend other dodgy rails. Delays drove passengers onto the airlines.

Low-cost airlines fill their planes differently from mainstream carriers. BA, British Midland, Air France and Lufthansa aim to make their money out of business travellers who pay over the odds to enjoy meals and loads of drinks in the air and on the ground in exclusive lounges. The economy seats are sold off, discounted as need be, some in advance and some at the last minute. Cheap seats are made available through downmarket travel agencies which publicise their deals through newspapers' classified columns.

The low-cost carriers see their aircraft as a series of buckets. The first set of buckets are the lowest-priced seats, with the eye-catching prices. Once these are all sold, demand flows into the next, slightly more expensive, bucket of seats. As the flight's departure approaches, seats get progressively more expensive. On a typical low-cost flight there could be up to ten different price buckets, with one-way fares ranging from £30 (\$42) to £210. But even the most expensive tickets tend to be cheaper than for the mainstream airlines.

Early assumptions that the low-cost carriers would struggle to make headway in the business market, because businessmen do not care how much their companies pay for their tickets, have turned out to be wrong. Stelios Haji-Ioannou, EasyJet's founder, says that one of the things he first noticed when the airline launched was how many business passengers he

seemed to be carrying. Not that business travellers are set apart, since everybody piles into the same no-frills cabin, with free-for-all seating and pay-for-all drinks and sandwiches; but business travellers tend to book late (and so more expensively) and travel mid-week. It turns out that businessmen are more price-sensitive than had been assumed. Some, presumably, are running their own businesses, so have an interest in keeping costs down; others are responding to cost-cutting memos from above.

WIRED FOR TAKE-OFF

The Internet has also helped the low-cost airlines. Airline tickets rival pornography as the hottest-selling commodity on the Internet, with sales estimated at more than \$5 billion worldwide. Mainstream airlines sell around 5% of tickets over the web. EasyJet decided to focus on Internet sales, so it offers discounts for online booking and has built a site that is easy to use. These days, some 90% of EasyJet bookings are made online. Ray Webster, EasyJet's chief executive, reckons that older or techno-illiterate people get a younger or more wired friend to do it for them.

Some 65% of Ryanair's bookings are made online. Even this figure is twice as high as the highest e-booking airline in America, Southwest, the original low-fare, no-frills carrier which was the model for the British low-cost operators.

The low-cost airlines have not just brought down the price of flying. They have changed the way British people travel, and also where they live, holiday and work. Air travel no longer involves the crowded hell of scheduled flights at Heathrow or charter flight delays at Gatwick. Cheap fares and European second homes have almost replaced house prices and school fees as a topic for dinner party chat.

At four o'clock on a summer afternoon at Luton airport, a queue is forming for the 5.40 to Edinburgh. A holidaying couple are returning to Edinburgh from Spain. The EasyJet flight was so cheap that it was worth their while taking an Airtours package from Luton, rather than flying from Scotland. Behind them is a management consultant who uses EasyJet from Luton because he lives just three exits up the M1 and it is quicker than hacking round the M25 to Heathrow. A technology transfer specialist at the Medical Research Council in Edinburgh says EasyJet is a way of coming down to London once a month for a fraction of the fare on British Airways or British

Midland. 'It seems silly paying extra when you are dealing with public funds,' she explains.

Businessmen from small and big companies alike are hopping onto cheap flights. Robert Jones, shipping and travel manager for Smit Land and Marine, a Liverpool-based Anglo-Dutch pipeline company, reckons he is saving £50,000 a year by using budget airlines. Half of his company's regular 30 trips a month from Liverpool to Amsterdam are by EasyJet, at £120 each, instead of the £350 he would spend on a scheduled airline. Travelling to Spain, he saves around £500 a trip. And, he says, the low-cost airlines make it easier to change passenger names if one employee has to substitute for another at a meeting. According to airport managers at Liverpool, since Ryanair and EasyJet have built up their flights from the city, the number of executive-type cars parked at the airport has shot up.

Leisure travel is changing too. Until recently, most people flew once or twice a year, to Florida or the south of Spain. These days, people increasingly hop on planes several times a year. It's no big deal any more. A salesman in a London electronics shop says his parents have a holiday house in the south of France, which he had stopped visiting after they stopped paying for his holidays. Recently, however, he has discovered that if he books ahead on the

Internet, he can fly EasyJet to Nice and back for under £50, making a monthly visit an affordable treat.

Perhaps the most astounding change is the number of long-distance commuters using the new airlines. Hang around long enough at Luton and you will meet a businessman or woman, usually middle-aged owners of companies in the south-east, who spend half their week as lotus eaters in Provence, nipping back for the other half to oversee their business being handled day-to-day by their staff. Most incredible of all, there is a resident of the Luton area who commutes to bustling Glasgow every morning, only to return to lovely Luton in the evening. That could be called some sort of progress.

Questions

- 1 What conditions make price discrimination possible in the airline industry, and what types of price discrimination are possible? Explain how different demand elasticities are relevant.
- 2 Explain the differences in pricing strategy between the mainstream airlines and the low-cost airlines, in terms of the different types of price discrimination used.
- 3 Explain the role of the Internet in the pricing strategies above.

10.5 Multiproduct Pricing

10.5.1 Context

In all the analysis of pricing so far it has been assumed that the firm is producing a single product. However, it was explained in Chapter 3 that this common assumption is not at all realistic; indeed it is difficult to think of many firms that do only produce a single product, while it is easy to think of firms that produce or sell thousands of different products. The reasons for making this assumption were also outlined in Chapter 3, and relate to the simplification of the analysis involved. It is true that there are some firms that produce or sell a wide diversity of products that are not related to each other in any way, either in terms of demand or production; in this situation the firm's pricing and output decisions can be examined using the same analytical framework as the single-product firm. However, this is a relatively rare situation. More often than not, firms producing multiple products face both demand and production interrelationships. It is now necessary to consider these relationships in more detail.

10.5.2 Demand interrelationships

Multiproduct firms frequently produce a product line or range of product lines, and many examples of this have already been discussed in previous chapters: cars, domestic appliances, athletic shoes, consumer electronics, and also services like banking, accounting and insurance. In some cases the products or models are substitutes for each other to some degree, which is usually the case within product lines, but in other cases the products are complementary, for example when Gillette produces shavers, blades and aftershave.

The consequence of such interrelationships is that a change in the price of one product affects the demand for others of the firm's products. If we consider a firm producing two products, X and Y, then the total revenue of the firm is given by:

$$TR = TR_X + TR_Y \quad (10.9)$$

where TR_X is the total revenue from product X, and TR_Y is the total revenue from product Y. The marginal revenues of each product are given by:

$$MR_X = \frac{\partial TR}{\partial Q_X} = \frac{\partial TR_X}{\partial Q_X} + \frac{\partial TR_Y}{\partial Q_X} \quad (10.10)$$

and

$$MR_Y = \frac{\partial TR}{\partial Q_Y} = \frac{\partial TR_X}{\partial Q_Y} + \frac{\partial TR_Y}{\partial Q_Y} \quad (10.11)$$

These equations show that when the quantity of one product sold by the firm changes, it not only affects the revenue obtained from the sale of that product, but it also affects the revenue obtained from the sale of other products of the firm. This is shown in the interaction term at the end of each expression; this term can be either positive or negative, depending on the nature of the relationship with the other product. If X and Y are in the same product line, and are therefore substitutes, the interaction term is negative, while if the products are complementary, as in the Gillette example, the interaction term is positive.

What are the implications of this interaction term? If firms ignore demand interdependencies they can make serious errors in decision-making. Take a firm like GM, with its various divisions. If the Chevrolet division cuts the price of its Camaro model, this will undoubtedly increase its sales and possibly revenues, if demand is elastic. The management of the Chevy division may regard this as a profitable exercise, if the increase in revenues exceeds any increase in costs. However, the corporate management at GM may take a different view if they look over at the Pontiac division, specifically the Pontiac Firebird, essentially a sister-car to the Camaro and a close substitute, and see its sales and revenues fall.

An opposite error can occur in the case of the Gillette example. A price cut in its shavers may not increase the revenue or profit from those shavers involved, and therefore such a price cut may be rejected as unprofitable. However, if the resulting increased sales in complementary products like blades and after-shave are considered, the price cut may be a profitable strategy for the firm as a whole.

10.5.3 *Production interrelationships*

When firms produce many products, and sometimes when they produce a single product, other products tend automatically to be generated at the same time because of the nature of the production process. These other products are often referred to as **by-products** or **joint products**. A good example is the production of petrol, which automatically involves the production of diesel oil and heating oil, among other products. Sometimes the resulting by-products are not desired; for example the production of many chemicals involves the creation of toxic substances and pollution. Since this often does not affect the firm's profit directly, this raises a public policy issue, which will be discussed in Chapter 12.

In many cases there are production interrelationships which are not inevitable but which are desirable. In Chapter 6 the concept of economies of scope was discussed; this referred to situations where it is less costly to produce two (or more) products together than to produce them separately. Such economies of scope are common in car production, the production of machinery and domestic appliances. Thus car manufacturing firms find it less costly to produce many different models, because they can share platforms and many production facilities. Better utilization of plant capacity becomes possible. The different models are not strictly joint products, but any pricing decision relating to one product must consider the effect on other products. For example, if Ford cuts the price of the Fiesta, the resulting increased sales may help to reduce unit costs of the Ford Focus, thus increasing profit through the **cost complementarity** between the two products.

10.5.4 *Joint products*

With joint products the production interrelationship is inevitable; when product X is produced, product Y will also be produced, whether this is desired or not. It is useful to classify such joint products into two main categories: those that are produced in fixed proportions and those that are produced in variable proportions.

a. Joint products produced in fixed proportions

This situation is easier to analyse because the products cannot be effectively separated from a production or cost standpoint, and therefore such products are not really multiple products at all, but are really product bundles.

Consider the following example:

SP10.2 Pricing of joint products

Procon PLC produces two products, pros and cons, in fixed proportions on a one-to-one basis, so that for every one pro produced one con is produced. The firm has the following total cost function:

$$C = 150 + 50Q + 2.5Q^2 \quad (10.12)$$

where Q is the number of product bundles produced, consisting of one pro and one con each. The demand functions for the two products are:

$$P_P = 200 - 2Q_P \quad (10.13)$$

$$P_C = 120 - 3Q_C \quad (10.14)$$

Where P_P and Q_P are the price (in £) and output of pros and P_C and Q_C are the price and output of cons.

Determine the optimal price and output for each product.

Solution

The above situation can be examined graphically or algebraically.

1. *Graphical analysis.* This is shown in Figure 10.3. This graph looks confusing at first, because it shows a lot of information. The two demand curves are labelled D_P and D_C , and the individual marginal revenue curves are shown as dashed lines and labelled MR_P and MR_C . The total or combined marginal revenue, MR , is kinked at the output of twenty units, since at that output, marginal revenue of cons becomes zero. Therefore beyond the output of twenty bundles the combined marginal revenue curve is the same as the MR curve for pros. The profit-maximizing equilibrium output is where $MR = MC$, and the corresponding prices for pros and cons can be obtained from the respective demand curves.

2. *Algebraic analysis.* Again this is based on the procedure in Chapter 8.

$$\begin{aligned} 1 \quad \text{Total revenue} &= R \\ &= (200 - 2Q_P)Q_P + (120 - 3Q_C)Q_C \\ &= 200Q_P - 2Q_P^2 + 120Q_C - 3Q_C^2 \end{aligned}$$

Now $Q = Q_P = Q_C$ since product bundles contain one pro and one con

$$R = 320Q - 5Q^2$$

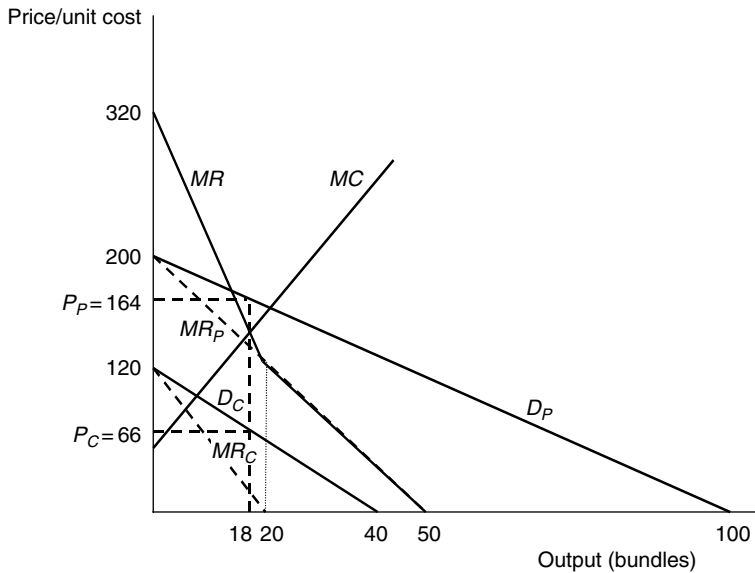


Figure 10.3. Optimal pricing for joint products produced in fixed proportions.

$$2 \quad MR = 320 - 10Q$$

$$3 \quad MR = MC$$

$$320 - 10Q = 50 + 5Q$$

$$-15Q = -270$$

$$Q = 18 \text{ units}$$

$$4 \quad P_P = 200 - 2(18) = \textbf{£164}$$

$$P_C = 120 - 3(18) = \textbf{£66}$$

The situation is made a little more complicated when the combined MR curve intersects the MC curve at an output larger than the level where the MR of cons becomes zero, which in the case of Figure 10.3 is twenty units. In this case the quantity of pros to be sold is still determined in the same way as before, but the quantity of cons sold will not exceed twenty. Any excess output must be thrown away, since releasing it onto the market will depress the price so much that profit will fall.

b. Joint products produced in variable proportions

This is again a more complicated situation, but, as usual, a more realistic one. An exact fixity of proportions is usually only observed in chemical reactions, when compounds are transformed into other substances in particular

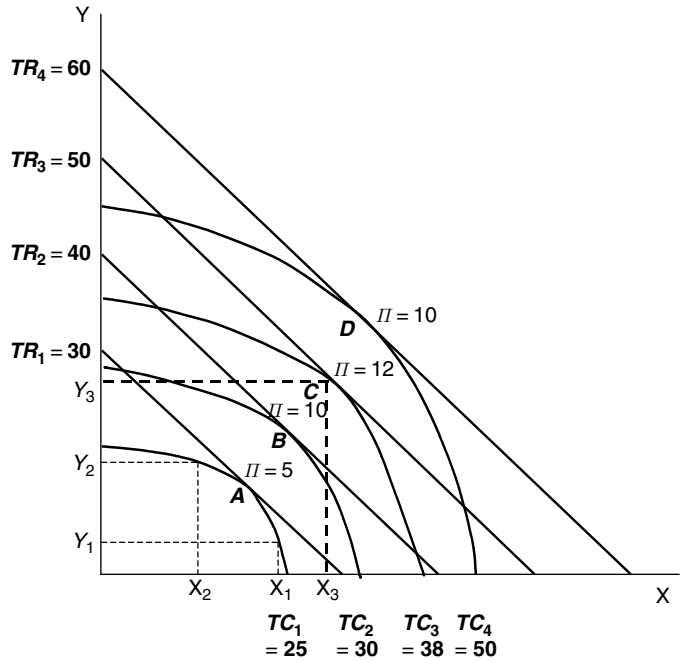


Figure 10.4. Optimal outputs of joints products produced in variable proportions.

quantities according to the laws of physics. Otherwise there is some flexibility in the processes involved that can increase or decrease the proportions according to profitability. The most common method of analysing this situation is to use a graphical approach, involving isocost and isorevenue curves. This is illustrated in Figure 10.4.

The concave (to the origin) curves on the graph are **isocost curves**: these curves represent **combinations of outputs which can be produced at the same total cost**. For example, TC_1 represents a total cost of 25 units; given this cost it is possible to produce X_1 units of X along with Y_1 units of Y, or X_2 units of X along with Y_2 units of Y, or any other combination of X and Y on the same curve. The isocost curves are shown as being concave to the origin, because it is assumed that there are diminishing returns in producing more of one product.

The sloping straight lines on the graph are **isorevenue curves**: these curves represent **combinations of outputs which result in the same total revenue**. These curves are shown as linear, which implicitly involves the assumption that the firm is a price-taker in each of the product markets. Points of tangency between isocost and isorevenue curves represent profit-maximization positions for any given cost or revenue. Thus point A on TC_1 and TR_1 yields more profit than any other point on TC_1 or TR_1 ; any other point on TC_1 will produce less revenue and therefore less profit, while any other point on TR_1 will involve more cost and therefore less profit. In order to find the overall profit-maximizing combination of outputs we have to find the point of

tangency with the highest profit; this occurs at point C, where combined profit from selling X and Y is 12 units. The optimal outputs of X and Y are therefore X_3 and Y_3 .

The reader may have observed that, while Figure 10.3 was concerned with the determination of optimal prices of joint products, Figure 10.4 was concerned with the determination of optimal outputs. The reason for this difference lies in the assumptions involved. There were a number of simplifying assumptions made in the model in Figure 10.4, for example only two products were involved. Most important in this context though is the assumption that the firm was a price-taker; thus the only relevant managerial decision was regarding the quantities of output. However, the price-taking assumption, which caused the total revenue curves to be linear, can easily be relaxed without affecting the analysis in any fundamental way. The profit-maximizing positions will still involve points of tangency between the total cost and total revenue curves. Once optimal outputs are determined, the prices of these outputs can be derived from the appropriate demand curves.

10.6 Transfer pricing

10.6.1 Context

As firms become larger there is a tendency for them to adapt a structure involving various divisions that in many cases are semi-autonomous. These divisions may be organized in various ways: they may perform parallel activities within the same areas, as is the case with GM; they may perform parallel activities in different areas or countries, as is the case with many multinationals; or they may perform vertically integrated activities in either the same or different areas. It is the last situation where transfer pricing is largely relevant. A transfer price represents ***an internal price within a firm, which is charged by one division to another for some intermediate product***. There are two main functions of charging such an internal price:

- 1 To foster greater efficiency within the firm as a whole, so that the whole firm will maximize profit.
- 2 To evaluate the performance of the various divisions of the firm, as profit centres in their own right.

As will be seen, these two functions can conflict with each other. Before considering an example it should be explained that transfer pricing can occur under three different conditions as far as the product is concerned: when there is no external market, when there is a perfectly competitive external market, and when there is an imperfectly competitive external market. These situations are now discussed in turn in conjunction with an example.

10.6.2 Products with no external market

Consider the following situation:

SP10.3 Transfer pricing

Due to the expansion of business in Gogoland in recent years, SR Products has set up a marketing division there responsible for selling a new product. The head office and manufacturing plant remain in the United States. The company has estimated that the total cost of manufacturing in the USA and the cost of transportation of the product is given by the function:

$$C = Q^2 + 20Q + 40 \quad (10.15)$$

where C = total cost per week in \$ and Q = units sold. This cost appears in the accounts of the manufacturing division.

The total cost for the marketing division in Gogoland is given by:

$$C = Q^2 + 140Q + 200 \quad (10.16)$$

This includes the \$100 per unit which is the transfer price paid by the marketing division to the manufacturing division. This total cost appears in the accounts of the marketing division. The revenue function for the marketing division is estimated as:

$$R = 500Q - 8Q^2 \quad (10.17)$$

where R = total revenue per week in \$.

- Calculate the optimal policy for the company as a whole, showing the price, output and overall profits of the firm.
- Calculate the optimal policies for the marketing division and the manufacturing divisions, assuming a transfer price of \$100, showing the overall profit of the firm in each case.
- Calculate the optimal transfer price in order to make the optimal policies for both divisions the same as that for the company as a whole.
- What does the above situation imply regarding managerial strategy?

As with all pricing problems discussed so far, the problem can be analysed either graphically or algebraically. In this case an algebraic approach will be used.

Solution

- Let us consider the firm as a whole to begin with. We can obtain the profit-maximizing position by finding the MC and MR for the firm as a

whole and equating them. First we can obtain the total cost to the firm by adding the costs of the manufacturing (N) and marketing (K) divisions:

$$TC = (Q^2 + 20Q + 40) + (Q^2 + 40Q + 200) = 2Q^2 + 60Q + 240 \quad (10.18)$$

Note that 100Q has been deducted from the costs of the marketing division, since this cost related to paying the transfer price to the manufacturing division, and was not a cost to the firm as a whole. Thus:

$$MC = 4Q + 60 \quad (10.19)$$

Total revenue is given by $500Q - 8Q^2$; thus:

$$MR = 500 - 16Q \quad (10.20)$$

Equating MC and MR:

$$4Q + 60 = 500 - 16Q$$

$$20Q = 440$$

$$Q = 22 \text{ units}$$

Price can be derived by substituting the value of Q into the demand equation; price is the same as average revenue, thus the demand equation is given by:

$$P = 500 - 8Q$$

$$P = 500 - 8(22) = \$324$$

This is the external price charged by the firm's marketing division.

The total profit of the firm can be obtained from its profit function:

$$\begin{aligned} \text{Profit} &= R - C = (500Q - 8Q^2) - (2Q^2 + 60Q + 240) \\ &= -10Q^2 + 440Q - 240 \\ &= \$4600 \end{aligned} \quad (10.21)$$

The above prices, output and profit are optimal for the firm as a whole.

b. However, the above result will not be achieved with the current strategy of charging a transfer price of \$100. Consider the manufacturing division: profit_N is maximized where $MR_N = MC_N$

$$MR_N = 100 \text{ (the current transfer price)}$$

$$MC_N = 20 + 2Q$$

$$\text{Equating: } 100 = 20 + 2Q$$

$$2Q = 80$$

$$Q = 40 \text{ units}$$

Given the firm's demand curve $P = 500 - 8Q$, this means that the firm will be able to sell this output at the price of **\$180**.

Total profit of the firm is obtained by substituting $Q = 40$ into the profit function in (10.21), giving a profit of **\$1,360**.

Now considering the marketing division:

Profit_K is maximized where $MR_K = MC_K$

$$MR_K = MR - MC_K = 500 - 16Q - (2Q + 40) = 460 - 18Q$$

$$MC_K = 100$$

$$\text{Equating: } 460 - 18Q = 100$$

$$18Q = 360$$

$$Q = 20 \text{ units}$$

Substituting this value of Q into the demand equation $P = 500 - 8Q$, the firm will charge the price of **\$340**.

Total profit of the firm is obtained by substituting $Q = 20$ into the profit function, giving a value of **\$4,560**.

c. In order to obtain the transfer price we need to consider the transferring division's marginal cost, that is MC_N . With no external market this represents the firm's optimal transfer price. The reason for this is that the firm wants to produce the profit-maximizing output for the firm as a whole, and the way to motivate the manufacturing (transferring) division to produce this output is to ensure that the division maximizes its profit at this output. Setting the transfer price, which is the manufacturing division's MR , equal to its MC at the overall profit-maximizing output ensures this. This condition can be seen more plainly by using the following analysis, which represents an alternative method of obtaining the optimal output for the firm, but which also shows how to compute the transfer price.

Profit maximization occurs when $MR = MC$

$$MR_K = MR - MC_K$$

$$MC_N = MC - MC_K$$

Therefore profit maximization occurs when

$$MR_K = MC_N \quad (10.22)$$

In the above example

$$MR_K = 500 - 16Q - (2Q + 40) = 460 - 18Q \quad (10.23)$$

$$\text{and } MC_N = 20 + 2Q$$

$$\text{Equating: } 460 - 18Q = 20 + 2Q$$

$$-20Q = -440$$

$$Q = 22 \text{ units (as before)}$$

The optimal transfer price (P_T) is therefore given by:

$$P_T = 20 + 2(22) = \mathbf{\$64}$$

Table 10.2. Implications of different transfer pricing strategies

	Transfer price	Market price	Output	Total profit	Profit of manufacturing division	Profit of marketing division
Firm	64	324	22	4,600	1,236	3,364
Manufacturing division	100	180	40	1,360	1,560	(200)
Marketing division	100	340	20	4,560	1,160	3,400

d. The results of the different transfer pricing strategies are shown in Table 10.2.

The situation above shows the importance to management of setting an optimal transfer price in order to maximize the profit of the firm as a whole. If the transfer price is too high, as here, the manufacturing (or transferring) division will produce too much, depressing the selling price and reducing overall profit. The marketing division would not want to buy as much as the optimal quantity, and again overall profit suffers.

10.6.3 Products with perfectly competitive external markets

Having considered the situation where there is no external market, it is now time to examine the situation where there is a perfectly competitive external market. In this situation the intermediate product can either be sold by the manufacturing division to another firm, or bought by the marketing division from another firm, at a fixed price. The optimal transfer price is the same as the external market price. If the transfer price were lower than this, it would be more profitable for the manufacturing division to sell the product externally, while if the transfer price were higher than this, it would be more profitable for the marketing division to buy the product externally.

One major difference between this situation and the one where there is no external market is that there may well be a mismatch between the amount that the manufacturing division wants to sell and the amount that the marketing division wants to buy. If the former exceeds the latter, then the excess can be sold on the open market, while if the latter exceeds the former, the excess can be purchased on the open market.

10.6.4 Products with imperfectly competitive external markets

This case is somewhat more complex than when external markets are perfectly competitive. The optimal transfer price this time equates the marginal cost of the manufacturing (transferring) division to the marginal revenue derived from the combined internal and external markets. The resulting optimal output is divided between internal transfers and external sales. The amount that is

transferred internally is determined by equating the marginal cost of the manufacturing division with the net marginal revenue from final product sales:

$$MC_N = MR_F - MC_K \quad (10.24)$$

The amount of the intermediate product that is sold in the external market is determined by equating the marginal cost of the manufacturing division with the marginal revenue from sales in the external market:

$$MC_N = MR_E \quad (10.25)$$

10.7 Pricing and the marketing mix*

So far we have ignored the other elements of the marketing mix in discussing pricing. Intuitively this would not appear to be sensible; if product quality is high, we would expect the price to be high, and a higher price might necessitate more spending on advertising and distribution. Thus we would generally expect there to be some interaction between the marketing mix variables. This interaction now needs to be examined.

10.7.1 An approach to marketing mix optimization

Let us assume that a firm has a marketing mix demand function of the following general form:

$$Q = f(P, L, A, D) \quad (10.26)$$

This is essentially (3.15) repeated. The firm's cost function can be expressed as follows:

$$C = g(Q, L)Q + A + D + F \quad (10.27)$$

where unit production cost $c = g(Q, L)$, meaning that unit cost is a function of output and product quality; A and D are discretionary costs of advertising (or promotion) and distribution, and F is a non-discretionary fixed cost. The profit function can now be expressed as follows:

$$\Pi = R - C = P f(P, L, A, D) - [g(Q, L)Q + A + D + F] \quad (10.28)$$

$$\Pi = P f(P, L, A, D) - g[f(P, L, A, D), L] f(P, L, A, D) - A - D - F \quad (10.29)$$

This clumsy-looking expression shows how much profit depends on all the different aspects of the marketing mix; both revenues and costs are affected by the levels of the marketing mix variables.

The necessary condition for finding the levels of these variables which optimize the marketing mix is obtained by partially differentiating the profit

function with respect to each of the variables and setting the partial derivatives equal to zero:

$$\frac{\partial \Pi}{\partial P} = \frac{\partial \Pi}{\partial L} = \frac{\partial \Pi}{\partial A} = \frac{\partial \Pi}{\partial D} = 0 \quad (10.30)$$

The mathematics involved in solving these equations is omitted for the sake of brevity, but can be found in an advanced marketing text, such as that by Kotler.⁹ The resulting conditions for optimization can be expressed in terms of the different elasticities, as follows:

$$\varepsilon_P = \frac{(PQ)}{A} \quad \varepsilon_A = \frac{(PQ)}{D} \quad \varepsilon_D = \frac{P}{c} \varepsilon_L \quad (10.31)$$

This is one form of the Dorfman–Steiner theorem,¹⁰ and it shows the relationships in optimality between the elasticities of the various marketing mix instruments. However, because the functions are only stated in general terms, the theorem does not directly give the optimal values of the variables. In order to see this more clearly we must be more specific regarding the form of the demand and cost functions, and this is the subject of the next subsection. We will then be able to see how an optimal ratio of advertising to sales revenue can be derived in terms of the ratio of the price and advertising elasticities (10.39).

In the following models, only three marketing mix instruments are considered: price, advertising and distribution. Product quality is omitted because its measurement is more complex, and in practice it is often estimated as a function of unit cost, as discussed in Chapter 3. Thus the concept ε_L in (10.31) relating to product quality elasticity can be understood as referring to the percentage change in demand caused by a 1 per cent change in quality, as measured in terms of unit cost.

10.7.2 The constant elasticity model

a. Nature of the model

It was seen in Chapter 3 that the constant elasticity demand function was in the power form:

$$Q = aP^b A^c D^d \quad (10.32)$$

whereas the linear model, featuring constant marginal effects, was in the form:

$$Q = a + bP + cA + dD \quad (10.33)$$

In these models, only three marketing mix instruments are considered: price, advertising and distribution. It was also explained in Chapter 3 that the power model is more realistic than the simpler linear model for two reasons:

- 1 It involves non-constant marginal effects. This allows for the existence of diminishing returns to marketing effort.

2 It involves interactions between the elements in the marketing mix. This means that marginal effects depend not only on the level of the variable itself but also on the values of the other marketing mix variables.

Thus in the linear model the marginal effect of advertising is given by:

$$\frac{\partial Q}{\partial A} = c$$

whereas in the power model the marginal effect is given by:

$$\frac{\partial Q}{\partial A} = caP^bA^{c-1}D^d = cQ/A \quad (10.34)$$

Since the value of Q is affected by the values of the other elements in the marketing mix, it can be seen that the marginal effect depends on the levels of these other elements.

We can now assume a linear cost function of the form:

$$C = uQ + A + D + F \quad (10.35)$$

where u represents a constant level of unit production cost. We can now apply the technique of partial differentiation of the profit function to obtain expressions for the optimal levels of price, advertising and distribution.

$$\begin{aligned} \Pi &= R - C = (P - u)Q - A - D - F \\ \frac{\partial \Pi}{\partial P} &= (P - u)\frac{\partial Q}{\partial P} + Q = 0 \\ \frac{\partial Q}{\partial P} &= bQ/P \text{ similar to (10.34)} \end{aligned}$$

Therefore

$$\begin{aligned} \frac{\partial \Pi}{\partial P} &= (P - u)bQ/P + Q = Q(b - bu/P + 1) = 0 \\ b - bu/P + 1 &= 0 \\ bu/P &= b + 1 \end{aligned}$$

$$P^* = bu(b + 1) \quad (10.36)$$

This expression is basically a rearrangement of expression (8.15) which was obtained in finding the optimal price and mark-up in terms of the price elasticity. However, we are now also in a position to find the optimal level of advertising and distribution in a similar manner:

$$\frac{\partial \Pi}{\partial A} = (P - u)\frac{\partial Q}{\partial A} - 1 = 0$$

Substituting (10.34):

$$\frac{\partial \Pi}{\partial A} = (P - u)\frac{cQ}{A} - 1 = 0$$

$$A^* = cQ(P - u) \quad (10.37)$$

$$\frac{\partial \Pi}{\partial D} = (P - u) \frac{\partial Q}{\partial D} - 1 = 0$$

Since

$$\frac{\partial Q}{\partial D} = dQ/D$$

similarly to (10.34):

$$\frac{\partial \Pi}{\partial D} = (P - u) \frac{dQ}{D} - 1 = 0$$

$$D^* = dQ(P - u) \quad (10.38)$$

It should be noted that the second-order conditions for optimality are not considered here, for reasons of brevity.

The question now is: how can these optimal levels of the marketing mix instruments be interpreted?

b. Interpretation of the model

Several interesting and not entirely intuitive conclusions arise from the above analysis. Assuming no interactions between firms, as discussed in the previous chapter, the following conclusions are the most important.

1. The optimal level of price is independent of the levels of the other marketing mix variables. This is in particular a surprising result, coming from expression (10.36). The explanation is that the other marketing mix variables, advertising and distribution, are essentially treated as sunk costs. However, it is possible that these other variables may be relevant in affecting the optimal price if they influence the price elasticity. This possibility is considered in the next subsection.

2. The optimal price appears to be independent of uncontrollable factors affecting demand. These include, for example, seasonal factors and the marketing mix of competitors. However, as in the above case, these other variables may affect the optimal price through their influence on price elasticity.

3. The optimal ratio of advertising to sales revenue can be calculated. This is performed by combining equation (10.36) and (10.37); the resulting ratio is given by:

$$\frac{A^*}{P^*Q} = -c/b = AED/PED \text{ in absolute terms} \quad (10.39)$$

This means that advertising and price should be set so that the resulting advertising-to-sales ratio is equal to the ratio of the advertising elasticity to

the price elasticity. The implication here is that firms should not simply use an arbitrary fixed ratio in order to determine advertising budgets; furthermore, the ratio should be adjusted if the firm suspects that either the advertising or price elasticities have changed.

4. The optimal ratio of advertising to distribution expenditure is equal to the ratios of their respective elasticities. This result is obtained by dividing (10.37) by (10.38), as shown below:

$$\frac{A^*}{D^*} = \frac{cQ(P-u)}{dQ(P-u)} = \frac{c}{d} \quad (10.40)$$

This means that if, for example, a firm's promotional elasticity is twice the level of its distributional elasticity, it should spend twice as much on promotion as on distribution. Again this is not entirely intuitive; some firms have reacted to having a higher elasticity by spending less on the relevant instrument, because they regard it as being unnecessary to spend so much in order to have the same effect.

10.7.3 Complex marketing mix interactions

The constant elasticity model is a very useful general model for representing the effects of the different elements of the marketing mix on sales and profit. However, managerial analysts may want to incorporate certain specific interactions into their demand models. A couple of examples will illustrate the situation.

- 1 It may be considered that the marginal effect of advertising is a function of distribution, because a greater number of retail outlets will increase the effects of a given advertising expenditure. A linear relationship may be involved as follows:

$$\frac{\partial Q}{\partial A} = eD \text{ where } e \text{ is a constant} \quad (10.41)$$

- 2 It may be considered that a greater level of advertising reduces price elasticity, by increasing brand loyalty. This may be modelled as follows:

$$PED = f/A \text{ where } f \text{ is a constant} \quad (10.42)$$

These relationships then have to be incorporated into the demand equation, according to its mathematical form. For example, if (10.42) is substituted into a constant elasticity demand function, we obtain:

$$Q = aP^{f/A}A^cD^d \quad (10.43)$$

The above examples just give an idea of the kind of interactions that can be incorporated into the marketing mix demand model. Obviously the equations become harder to work with mathematically, and may cause problems in

regression analysis, but they may lead to more reliable and useful results, both in terms of testing economic theories, and in terms of making better managerial decisions.

10.8 Dynamic aspects of pricing

10.8.1 *Significance of the product life-cycle*

This refers to the concept that a firm will tend to adopt different pricing practices for a product at different stages of its product life-cycle. This area of pricing strategy has been somewhat neglected in many economics and marketing texts, and indeed in research.¹¹ However, it is apparent when we consider many products that their prices have changed significantly since the date they were originally introduced in the market. In some cases prices have risen considerably, while in other cases they have dropped significantly. The reasons for these changes and differences need to be examined.

It was stated at the beginning of the chapter that the pricing decision was generally not the most fundamental one that management has to take. The positioning and product decisions tend to be paramount, but, as will be seen in the next chapter, these are long-run decisions and require a long-run frame of analysis. They also involve an interdependence of marketing mix instruments, in particular the interdependence between product characteristics and price. Therefore, any discussion of pricing strategy should take into consideration both this interdependence and the fact that a product's price should normally change during the course of the product life-cycle. The interdependence aspects are discussed more fully in the next section; this section is concerned with the relationships between pricing strategy and the product life-cycle.

The long-run decisions regarding whether to produce a particular product or not involve a discussion of investment analysis, which is the topic of the next chapter. Without anticipating this discussion in detail, it involves an examination of profits or cash flows over the whole lifetime of a project. This in turn means that future prices of the product have to be estimated. Since demand and cost factors change over a product's life-cycle it follows that the product's price is likely to change during the course of the cycle. It is of great importance to management to be able to estimate these changes as accurately as possible before launching the product in order to make the initial fundamental decision on whether to produce the product.

10.8.2 *Early stages of the product life-cycle*

The evidence available,¹²⁻¹⁴ tends to suggest that price elasticity can be expected to decrease for the first three phases of the cycle: introduction, growth and maturity. At the introduction stage there is normally much expenditure on promotion to gain recognition and awareness; discounting, coupons and free samples are common. This means that a **market penetration**

strategy is often advantageous, because demand is highly elastic. As the product gains in image and brand loyalty, and product differentiation increases, demand becomes less elastic, and prices are generally raised.

Of course, it is easy to think of products where the opposite has occurred. This applies in particular to high-technology consumer durables, like VCRs, microwaves and mobile phones. Consumer behaviour in this situation involves innovators and early adopters being willing to pay high prices to own a product that has some prestige value in terms of being novel and exclusive. Thus demand tends to be less elastic and a **market skimming** strategy is advantageous. In this case the price can fall considerably as the product passes through the introduction and growth stages. Competition springs up, often with better or more advanced products, and unit costs fall on account of economies of scale and learning curve effects.

10.8.3 Later stages of the product life-cycle

Once a product has reached maturity there is usually a considerable amount of competition in the market. Emphasis tends to switch from product innovation to process innovation; products become more standardized and cost minimization becomes an important factor. At this point, evidence suggests that demand elasticity increases again, as the product moves into the decline phase. Curry and Riesz¹⁵ have found that the mean price of all brands within a product form tends to decline over time (net of inflation), and that the price variance within a designated product form also tends to decline over time. This is again what one would expect with an increase in competition and the availability of close substitutes. Curry and Riesz suggested that 'Price, which previously may have been a real or fictitious surrogate for product quality, gradually loses its flexibility as both a strategic and functional marketing variable.' This conclusion appears to ignore the promotional potential in pricing. This aspect and the relationship between price and product quality are considered in the next section.

10.9 Other pricing strategies

This section is rather miscellaneous in nature; it discusses a number of other pricing strategies that are found in practice, but which are more difficult to model in economic terms. This does not mean that these strategies do not help the firm to maximize profit in the short or long run, but that they tend to relate to more complex aspects of consumer behaviour that are not generally taken into consideration in the neoclassical model. It is more appropriate to refer to these models as **behavioural models**.

These models generally try to identify the key factors that determine consumer buyer behaviour. Zeithaml's means-end model is a good example;¹⁶ this examines perceived quality, perceived price, the price-quality relationship

and perceived value. These factors are now considered in turn, along with various pricing strategies that are based on them.

10.9.1 *Perceived quality*

This concept is a subjective assessment, similar to an attitude, not an objective factor relating to intrinsic physical attributes, although obviously the latter serve as cues from which consumers often infer quality. Thus perceived quality represents an abstract concept of a global nature, and judgements made regarding it are made in a comparative context. Other non-physical cues are extrinsic; these include brand image,¹⁷ store image,¹⁸ advertising level,¹⁹ warranty, and of course price, which is examined in more detail shortly. These cues tend to be more important in purchase situations when intrinsic cues are not available and when there is more perceived risk,²⁰ for example many services, and when quality is difficult to evaluate, as with experience goods, i.e. goods which have to be experienced before the consumer can evaluate the quality.

10.9.2 *Perceived price*

The main point here is that the perceived price may be different from the actual price for a number of reasons, some of which have been mentioned in previous chapters. For example, consumers consider search, time and psychic costs as being part of the perceived price.²¹ This means that when they encode price information, they include these additional costs.

Evidence suggests that for certain goods, price information is not encoded at all by a substantial proportion of consumers. Dickson and Sawyer²² reported that for four types of product, margarine, cereal, toothpaste and coffee, 40 per cent or more of consumers did not check price. Most of these consumers said that price was just not important in the purchase decision. Clearly the products mentioned are all fast-moving consumer goods (FMCG) and one might expect that in this situation price would not be an important decision factor; in the purchase of a stereo or holiday one might expect different behaviour.

10.9.3 *The price-quality relationship*

This aspect of consumer behaviour has been very well researched. However, the findings are difficult to summarize in any brief manner, since there is a considerable amount of conflicting evidence. It appears that there are various other quality cues that are more important, like brand name and store image;²³ also studies show that price is only weakly correlated with objective quality.^{24, 25} However, there do appear to be some situations where price is used as an indicator of quality: when other cues are absent, and when there is much price or product quality variation within a product class.

There is certainly evidence that a high price can give prestige to a product, by making it exclusive, even if the evidence of greater objective quality is

dubious. This leads to the strategy sometimes referred to as '**prestige pricing**'; German luxury saloon and sports cars, like Mercedes, BMW and Porsche, enjoy this cachet. This is not to imply that such cars are lacking in objective quality! However, other firms may try to employ the same strategy without necessarily ensuring a higher-quality product.

Some researchers have found evidence that consumers have a range of acceptable prices for a product, with too high a price being seen as too expensive, and too low a price indicating inferior quality.^{26, 27} This can help to explain the phenomenon of '**odd pricing**', a universally observed pricing strategy. An odd price usually ends in a '9', for example £399. This would be acceptable for a consumer whose price range was £300 to less than £400, which is a common way of defining a price range. Research shows that consumers perceive a much greater price difference between £399 and £400 than between £398 and £399.

Another strategy based on this same concept of acceptable price ranges is '**price lining**'. Instead of starting with a positioning and product concept, a firm starts with a concept of a particular price range which appears to represent a gap in the market, in terms of there being consumer demand in this range but a lack of products currently available in the relevant range. A product is then identified with characteristics that would cause it to be priced accordingly. Japanese consumer electronics firms have used this strategy successfully. Profit maximization can still be the objective, but the normal order of decision-making is reversed.

10.9.4 *Perceived value*

Consumers can interpret this concept in different ways. To some consumers it simply means a low price. However, in the majority of cases perceived value is a function of the ratio of perceived quality to perceived price, and this concept has been incorporated into Keon's bargain value model.²⁸ Some empirical evidence supports this model as being a good indicator of probability of purchase.²⁹

Many **price promotion** strategies are based on aspects of perceived value. These strategies all involve some kind of discounting. Evidence suggests that these discounts tend to be more effective when **reference prices** are used,^{30, 31} for example 'normal price £599, now only £499'. The reference price can be used to persuade consumers that quality is high relative to the current price being charged, implying that they are being offered good value.

Summary

- 1 Pricing is only one component of the marketing mix and pricing decisions should be interdependent with other marketing mix and positioning decisions.

- 2 Competitive advantage refers to the situation where a firm creates more value than its competitors, where value is measured in terms of perceived benefit minus input cost.
- 3 The value created by a firm can be divided into consumer surplus and producer surplus; the former represents the excess of perceived benefit over the price paid, while the latter represents the excess of the price charged over the input cost.
- 4 Market positioning essentially involves aiming for a cost advantage or a benefit advantage.
- 5 Positioning depends both on the nature of a firm's competitive advantage, in terms of its resources and capabilities, and on environmental forces in the market, particularly those relating to customers and competition.
- 6 Price elasticity is important in guiding a firm's general pricing strategy, in terms of aiming for increasing market share or increasing profit margin.
- 7 Segmentation involves dividing a market into component parts according to relevant characteristics related to buyer behaviour.
- 8 Segmentation and targeting are important because different strategies are appropriate for different segments.
- 9 Targeting involves determining whether a broad coverage strategy is appropriate, or whether a focus strategy is better. In the latter case the appropriate product or segment must be selected, again according to competitive advantage.
- 10 Price discrimination means charging different prices for the same or similar products, where any price differentials are not based on differences in marginal cost.
- 11 Price discrimination always increases the profit of the seller because it enables the seller to capture some of the consumer surplus.
- 12 Price discrimination can only occur if market segments have different demand elasticities and they can be separated from each other.
- 13 Firms producing many products in a product line or product mix face more complex pricing decisions because of demand and cost interdependencies.
- 14 Transfer pricing occurs when one part of a firm charges an internal price to another part of the same firm, a common practice in large firms.
- 15 Charging the right transfer price is important to the firm, not just in maximizing overall profit, but also in evaluating the performance of different divisions.
- 16 Pricing is only one element in the marketing mix, and pricing decisions need to be made in conjunction with other marketing mix decisions.
- 17 Interactions between different components of the marketing mix need to be carefully considered by managers when constructing the demand models necessary for making pricing and other marketing mix decisions.
- 18 Firms generally charge different prices for a product during different stages of the product life-cycle, even if there are no changes in quality. These changes have to do with changes in demand elasticity, and also often with unit cost changes.

- 19 Behavioural models are important in that they can enable managers to understand consumer behaviour and reactions to the firm's marketing strategies. Pricing strategies therefore need to take these models into consideration, even though they are more complex than the traditional neoclassical economic model of consumer behaviour.

Review questions

- 1 Explain why it is important for managers to know the principles of price discrimination.
- 2 Explain the relationship between a product line and joint products.
- 3 Assuming that there is no external market for the product, how should a firm determine the optimal transfer price for an intermediate product?
- 4 Explain how the price elasticity for a product is likely to change during the product life-cycle.
- 5 Explain the meaning and significance of the concept of perceived value. How is it related to the strategy of prestige pricing?
- 6 Why is it not sufficient for a firm to create value in order for it to have a competitive advantage?
- 7 How does branding relate to competitive advantage? Why do all firms not brand their goods, if this enables them to raise their price?
- 8 Explain why providers of mobile phone services should segment their markets. What criteria are relevant for segmentation in this situation?
- 9 Would you expect an airline flying a transatlantic route to pursue a broad coverage or a focus strategy? What factors would affect this decision?

Problems

- 10.1 GMG, a cinema complex, is considering charging a different price for the afternoon showings of its films compared with the evening ticket price for the same films. It has estimated its afternoon and evening demand functions to be:

$$P_A = 8.5 - 0.25Q_A$$

$$P_E = 12.5 - 0.4Q_E$$

where P_A and P_E are ticket prices (in £) and Q_A and Q_E are number of customers per week (in hundreds). GMG has estimated that its fixed costs are £2,000 per week, and that its variable costs are 50 pence per customer.

- a. Calculate the price that GMG should charge if it does not use price discrimination, assuming its objective is to maximize profit.
- b. Calculate the prices that GMG should charge if it does use price discrimination.

- c. Calculate the price elasticities of demand in the case of price discrimination.
- d. How much difference does price discrimination make to profit?

10.2 TT Products produces two items, bibs and bobs, in a process that makes them joint products. For every bib produced two bobs are produced. The demand functions for the two products are:

$$\text{Bibs: } P = 40 - 4Q$$

$$\text{Bobs: } P = 60 - 3Q$$

where P is price in £ and Q is units of each product. The total cost function is:

$$C = 80 + 20Q + 4Q^2$$

where Q represents a product bundle consisting of one bib and two bobs.

- a. Calculate the prices and outputs of bibs and bobs that maximizes profit.
- b. Calculate the size of the above profit.

10.3 DPC is a firm that has separate manufacturing and marketing divisions. The cost functions of these divisions are as follows:
Manufacturing:

$$C = 2Q^2 + 5Q + 1.5$$

Marketing:

$$C = Q^2 + 3Q + 0.5$$

where C is total cost (in £ million) and Q is output (in millions of units per year). The firm's demand function for its final product has been estimated as:

$$P = 20 - 5Q$$

where P is the price of the final product (in £).

- a. Calculate the profit-maximizing price and output of the final product.
- b. Calculate the optimal transfer price.
- c. Calculate the effect on profit if the transfer price is £6.
- d. Calculate the effect on profit if the transfer price is £10.

10.4 Gungho Products has estimated the demand function for its new soft drink to be:

$$Q = 320P^{-1.5}A^{0.4}D^{0.2}$$

where Q is measured in cans sold per month, P is the price in £, A is advertising expenditures in £ per month and D is distribution expenditures

in £ per month. Unit production costs are estimated at £0.25 per can and are constant over the firm's output range. The firm has fixed costs of £30,000 per month.

- a. Determine the firm's optimal price.
- b. Determine the firm's optimal advertising and distribution expenditures.
- c. Comment on the relationship between these two expenditures.
- d. Determine the optimal advertising-to-sales ratio for the firm.
- e. Determine the firm's level of profit.

Notes

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